

Probability and Its Associated Calculus

UChicago Political Science Math Camp

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2025

bayesian (1).jpg

Probability

Probability is the **likelihood of an event occurring**

We denote the probability of an event A as a real number between 0 and 1:

$$\Pr[A] \in [0, 1]$$

Discrete vs. Continuous Probability

There are two “categories” in studying probability:

1. **Discrete** probability, where there are $n \in \mathbb{N}$ possible outcomes or events (“What number will this dice roll?”)
2. **Continuous** probability, where there are $n \rightarrow \infty$ possible outcomes or events (“How far will I throw this dice?”)

This refers to the *measurement*, not the *probability* itself

Axioms of Probability

There are three basic axioms when working with probability:

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3.

$$a_i \cap a_{-i} = \emptyset \quad \forall a_i, a_{-i} \iff \Pr \left[\bigcup_{i=1}^k a_i \right] = \sum_{i=1}^k \Pr[a_i]$$

Applying Axioms

Show these axioms for each of the following discrete probabilities:

1. A fair two-sided coin
2. An unfair six-sided dice
3. Tomorrow's temperature (in whole degrees, ok to place bounds)

Basic Probability

Let there be an event with equally likely outcomes $A_{i \in n} \in \{S\}$. Then, the probability that a draw yields an outcome meeting some condition Q is:

$$\Pr[a_i \rightarrow Q] = \frac{|a_i \rightarrow Q|}{|S|}$$

Count of “target” outcomes over count of total

Rolling a Dice

Suppose we roll a (fair) 100-sided dice to recover $a \in \{x \in \mathbb{N} : 1 \leq x \leq 100\}$. What is the probability of each outcome?

1. $a = 15$

2. $a \leq 50$

3. $a < 50$

4. a is even

5. $\frac{a}{3} \in \mathbb{N}$

6. $\frac{a}{7} \in \mathbb{Q}$

Combinatorics

To determine $|S|$, we often rely on combinatorial mathematics, and use the formula:

$$C = \binom{n}{a} = \frac{n!}{a!(n-a)!}$$

Where

- ▶ C = number of unique combinations (outcomes)
- ▶ n = number of objects in a set
- ▶ a = number of objects chosen from the set

Relaxed Assumptions

Of course, not all events are “fairly” distributed:

- ▶ A cat landing on its feet
- ▶ An authoritarian election (mechanism design)
- ▶ Getting pulled over while speeding

Often, we can observe the outcomes, but must estimate its true probability

Independence

Two events are said to be **independent** or **disjoint** if the probability of one is not tied to the probability of the other:

- ▶ Metra running late and the White Sox score
- ▶ Math background and field seminar grades

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Two events are **dependent** if the expected outcome of one *is* affected by the outcome of the other:

- ▶ Delays on the 171 and delays on the 172
- ▶ Numbered street and apartment rent

Conditional Probability

Many events are contingent on prerequisite outcomes. The **conditional probability** of an outcome A given an event B can be expressed as:

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}$$

A Note on Dependency

A dependent relationship can exist in different mechanisms:

- ▶ **Correlational:** The outcome of one event does not directly impact another, but signals underlying conditions
- ▶ **Causal:** The outcome of one event is causally related to the outcome of another

For now, our quantitative mechanisms are the same for each.

Replacement

One common example of independence versus dependence is whether or not an outcome is “replaced” after a random draw:

- ▶ A card is drawn from a deck, then shuffled back in → **independent**
- ▶ A card is drawn from a deck, then discarded → **dependent**

Why might this be?

I.I.D.

Independent and identically distributed:

- ▶ Independent: The expected outcome of the n th event is not impacted by any $n - k$ th outcome.
- ▶ Identically Distributed: Every draw is identical, and occurs with full replacement

Joint Probability

For two independent and disjoint events A and B , their **joint probabilities** are as follows:

$$\Pr[A \cap B] = \Pr[A] \cdot \Pr[B]$$

$$\Pr[A \cup B] = \Pr[A] + \Pr[B]$$

How might this change for non-disjoint events?

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How might this change for non-disjoint events?

$$\implies \Pr[A \cup B] = \Pr[A \cup \neg B] + \Pr[\neg A \cup B] + \Pr[A \cap B]$$

Joint Probability Example

We want to evaluate each quadrant for $\Pr[i \cap j]$:

		j	
		Sunny	Cloudy
i	Rainy		
	Dry		

Expanding Conditional Probability

Suppose...



Bayesian Statistics

Suppose we know the probability of A given B , but we want to know the probability of B given A . We can use **Bayes' Rule**:

$$\Pr[A \mid B] = \frac{\Pr[B \mid A] \cdot \Pr[A]}{\Pr[B]}$$

Random Variables

A random variable is a function X such that there exists a real number for every $s \in S$:

$$X : S \rightarrow \mathbb{R}$$

There is then some non-deterministic probability $\Pr[X(s) = x] \in [0, 1]$ for every event s .

PMFs

For a discrete random variable X , we can model a **probability mass function**:

$$\text{PMF} \equiv f(x) = \Pr[X = x]$$

This models the likelihood of any given outcome of $x \in S$.

What do our probability axioms imply about PMFs?

PMF examples

How might we construct the PMFs of the following?

- ▶ A fair, six-sided dice
- ▶ An unfair six-sided dice (twice as likely to land on 6)
- ▶ A nation's global ranking in GDP (assume $n = 200$ for simplicity)

CDFs

We often want to represent the probability that a random variable is larger or smaller than some threshold. For this, we can model a **cumulative distribution function**:

$$\text{CDF} \equiv F(x) = \Pr[X \leq x] = \sum_{i \leq x} \Pr[i]$$

What do our probability axioms imply about discrete CDFs? How might we solve $\Pr[X \geq x]$?

CDF examples

How might we construct CDFs of the following?

- ▶ A fair dice rolling at least a 3
- ▶ A nation being in the top n th percentile globally of GDP (assume $N = 200$ for simplicity)
- ▶ The “amount” of persuasion voters need to choose candidate A in a two-race, majoritarian election

Uniform Distribution

The **uniform distribution** models every outcome as equally likely:

$$f(x) = \Pr[X = x] = \frac{1}{|X|}$$

Note that $f(x)$ is a *constant function* – what does this imply?

Another Important Discrete Distribution

Suppose you flip a fair coin twice. What is the probability of recovering H at least once?

Suppose you roll a fair dice six times. What is the probability of recovering 6 at least once?

A Closing Derivation

Let each draw of an event consist of n equally likely outcomes across N alternatives. What is the probability of drawing some outcome $n \equiv Q$ over the period of draws $1, 2, \dots, t$ for a condition Q ?

The Binomial Distribution

To model the likelihood of an event of probability p occurring within n i.i.d. trials, we can use the **binomial distribution**:

$$B(n, p) = \binom{n}{k} p^k q^{n-k}$$

Where $\binom{n}{k}$ is the **binomial coefficient** for k outcomes over n trials

Expectation (A Minor Recap)

The **expected value** of a random variable is the sum of its possible outcome values weighted by the probability that value occurs:

$$E[X] = \sum_x \Pr[x] \cdot x$$

With discrete RVs, this summation is simple and well-defined, even if not always meaningful

Expectation Examples

Calculate the expected value of the following:

1. A fair six-sided dice roll
2. The sum of two fair six-sided dice rolls
3. The **maximal value** of two fair dice rolls

Extending Expectation

Let's evaluate this. Google "Roll Two Dice", and record 10 iterated outcomes each.

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What do we start to see when plotting these outcomes?

And More So, Large N

What if we could simulate a more sophisticated result – the maximal value of three dice rolls?